

DaSEH Day 3 Cheatsheet

Subsetting Data in R

Functions

Library/Package	Code	Example	What it does
Base R	<code>nrow(); ncol()</code>	<code>nrow(x); ncol(x)</code>	Get the number of rows and columns in object <code>x</code> , respectively.
Base R	<code>dim()</code>	<code>dim(x)</code>	Get the number of rows and columns in object <code>x</code> .
tidyverse (dplyr)	<code>glimpse()</code>	<code>glimpse(mtcars)</code>	Get an overview of data frame <code>x</code> .
tidyverse (dplyr)	<code>slice_sample()</code>	<code>slice_sample(mtcars)</code>	View a random subset of rows from <code>x</code> .
Base R	<code>data.frame()</code>	<code>df <- data.frame(1:3)</code>	Create a data frame. Named arguments must have the same length.
tidyverse (tibble)	<code>tibble()</code>	<code>tibble(mtcars)</code>	Create a tibble from a data frame or matrix.
tidyverse (tibble)	<code>rownames_to_column()</code>	<code>df <- rownames_to_column(df, "new_col")</code>	Convert row names to a new first column. The argument names the new column.
Base R	<code>colnames()</code>	<code>colnames(df)</code>	Get or set column names for a matrix or data frame.
tidyverse (dplyr)	<code>rename()</code>	<code>df <- rename(df, MPG = mpg)</code>	Rename selected columns while keeping all variables.
tidyverse (dplyr)	<code>rename_with()</code>	<code>df <- rename_with(df, toupper)</code>	Rename selected columns with a function.
Base R	<code>toupper()</code>	<code>toupper("text")</code>	Convert lowercase characters to uppercase.

Library/Package	Code	Example	What it does
Base R	<code>tolower()</code>	<code>tolower("TEXT")</code>	Convert uppercase characters to lowercase.
janitor	<code>clean_names()</code>	<code>clean_names(df)</code>	Return a data frame with clean names.
tidyverse (dplyr)	<code>pull()</code>	<code>pull(df, "existing_variable_name")</code>	Extract a column as a vector. Base R notation: <code>df\$column</code> .
tidyverse (dplyr)	<code>select()</code>	<code>select(df, "existing_variable_name")</code>	Select columns that match the specified argument. (<i>See note</i>)
tidyverse (dplyr)	<code>filter()</code>	<code>filter(df, mpg > 20)</code>	Return rows that match the specified logical condition.
Base R	<code>==, <=, >=, !=</code>	<code>filter(df, mpg > 20)</code>	Compare values in an object. Useful with <code>filter()</code> .
Base R	<code>%in%</code>	<code>filter(df, mpg %in% c(20, 21, 22))</code>	Check whether values on the left are in the object on the right. Returns TRUE or FALSE.
Base R	<code> ></code>	<code>df <- df > select("variable_name")</code>	Pipe a data frame through additional operations.
tidyverse (magrittr)	<code>%>%</code>	<code>df <- df %>% select("variable_name")</code>	Pipe a data frame through tidyverse operations.
tidyverse (dplyr)	<code>mutate()</code>	<code>df <- mutate(df, newcol = wt / 2.2)</code>	Add a new column based on existing columns.
tidyverse (dplyr)	<code>relocate()</code>	<code>df_carb <- relocate(.data = df, wt, .before = mpg)</code>	Reorder columns in a data frame or tibble.
tidyverse (dplyr)	<code>arrange()</code>	<code>df <- arrange(df, mpg)</code>	Reorder rows in ascending order. Use <code>arrange(desc())</code> for descending order.

NOTE: See [tidyselect helpers](#) (`?dplyr_tidy_select`) for handy logical functions to use with `select()`.

* This cheatsheet format was adapted from Alex's Lemonade Stand materials ([source](#)).

* Find more resources at <https://daseh.org>.